LESSON 4.1

NEGATIVE EXPONENT LAW

LESSON INTRODUCTION

MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

LEARNING TARGETS

- I can apply the laws of exponents to evaluate numerical expressions with integer exponents.
- I can generate equivalent numerical expressions using the laws of exponents for expressions with integer exponents.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational bases.

WORKING TOWARD

MA.912.NSO.1.1 Extend previous understanding of the laws of exponents to include rational exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

ESSENTIAL QUESTIONS

- How do you evaluate numerical expressions that have negative exponents?
- What patterns exist when evaluating numerical expressions with integer bases and exponents?

LESSON NARRATIVE

Students extend their prior knowledge of the laws of exponents to be able to work with numeric expressions having negative exponents. Students explore patterns in the values of expressions having the same base and different exponents. Students discover how to write equivalent expressions using positive exponents and use the laws of exponents to evaluate expressions.

COMPONENT SUMMARY

4.1.1 WARM-UP *(~5 MIN.)*

Ichthyologist career profile

4.1.2 REFRESHER *(5 MIN.)*

Review laws of exponents

4.1.3 EXPLORATION *(15 MIN.)*

Explore patterns in expressions with positive and negative exponents

4.1.4 BRING IT TOGETHER *(5 MIN.)*

Solidify understanding with negative exponent law

4.1.5 GUIDED PRACTICE *(5-10 MIN.)*

Rewrite expressions using positive exponents

4.1.6 COLLABORATION *(10-15 MIN.)*

Apply other exponent laws to rewrite expressions with negative exponents

4.1.7 WRAP-UP *(~5 MIN)*

Reflect on mastery of learning targets

RESOURCES

GLOSSARY

Key Terms:

- Negative exponents
- Negative exponent law

Additional Terms:

- Laws of exponents

MATERIALS

- Career Profile video
- (Optional) Highlighters

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not understand how to apply the law of negative exponents when given a fraction with a negative exponent in the denominator.
- Students may believe that a base with an exponent of zero equals zero.
- Students may confuse the concepts of the “opposite” and the “reciprocal.”
- Students may struggle to understand why dividing by a number is the same as multiplying by the reciprocal of the number.

THINGS TO CONSIDER

- As students progress through the lesson, consider prompting them to create a reference sheet (or foldable) with the laws of exponents. There are several laws reviewed, and students may benefit from seeing examples of each law on a reference sheet or in a foldable.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

4.1.3 Exploration prompts students to compare and connect the patterns they see as the base stays the same and the exponent changes. In questions 5 and 6, students compare and connect the patterns they see when the base and exponent are the same and the sign of the exponent is different. Students apply their understanding to answer question 8.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

4.1.6 Collaboration provides students the opportunity to apply the laws of exponents to numerical expressions that include negative exponents and/or bases. Students discover that the laws of exponents hold for numeric expressions that have negative exponents.

DIFFERENTIATE INSTRUCTION

4.1.6 Collaboration provides an opportunity for auditory learners to engage in discussion as they work through the process of applying the laws of exponents to numerical expressions with a partner. 4.1.6 Collaboration also allows visual learners to see an expression in different but equivalent forms before finding the value.

$6^{-2} \cdot 6^5$	$\frac{1}{6^2} \cdot 6^5$	$\frac{1}{6 \cdot 6} \cdot 6 \cdot 6 \cdot 6 \cdot 6 \cdot 6$	$\frac{1}{36} \cdot 7776 = 216$
$(-4)^3 \cdot (-4)^{-7}$	$-4^3 \cdot \frac{1}{4^7}$	$\frac{-4 \cdot -4 \cdot -4 \cdot 1}{4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4}$	$\frac{-64}{16384} = \frac{-1}{256}$
$3^{-4} \cdot 3^4$	$\frac{1}{3^4} \cdot 3^4$	$\frac{1}{3 \cdot 3 \cdot 3 \cdot 3} \cdot 3 \cdot 3 \cdot 3 \cdot 3$	$\frac{1}{81} \cdot 81 = 1$

4.1.1 WARM-UP: DAY AT WORK WITH A FISH BIOLOGIST

TEACHER MOVES

Have students write something they are passionate about on sticky notes and collect them. This information could be useful for a variety of uses at the teacher level throughout the year.

4.1.1 Warm-Up: Day at Work with a Fish Biologist

Moises, an ichthyologist, or fish biologist, says, “Try to be in touch with a system that you really want to study, with a system that really drives you crazy, because that’s how you nurture passion and passion is one of the most important things in becoming a researcher.”

1. What is something that you are passionate about?
   Answers will vary. Something that I am passionate about is playing sports.

2. Why do you think he says that passion is one of the most important things in becoming a researcher?
   Answers will vary. Passion is very important because despite the ups and downs in every journey, your enthusiasm behind your passion keeps you going.

4.1.2 Refresher: Laws of Exponents

1. Complete the rule for each exponent law shown.

Exponent Law	Numerical Example	Algebraic Rule
Product of Powers	$2^3 \cdot 2^1 = 2^4$	$a^m \cdot a^n = a^{m+n}$
Power of a Product	$(-7 \cdot 3)^2 = (-7)^2 \cdot 3^2$	$(ab)^m = a^m \cdot b^m$
Power of a Power	$(6^3)^4 = 6^{12}$	$(a^m)^n = a^{m \cdot n}$
Quotient of Powers	$\frac{(-3)^7}{(-3)^2} = (-3)^5$	$\frac{a^m}{a^n} = a^{m-n}$
Power of a Quotient	$\left(\frac{13}{5}\right)^3 = \frac{13^3}{5^3}$	$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$
Identity Exponent	$15^1 = 15$	$a^1 = a$
Zero Exponent	$127^0 = 1$	$a^0 = 1$

2. Write an equivalent exponential expression by applying the laws of exponents.

a. $\frac{11^{14}}{11^8} =$
$11^{14-8} = 11^6$

b. $10^4 \cdot 10^3 =$
$10^{4+3} = 10^7$

c. $5^2 \cdot 5^{10} =$
$5^{2+10} = 5^{12}$

d. $(4 \cdot -5)^2 =$
$4^2 \cdot (-5)^2$

e. $(2^3)^8 =$
$2^{3 \cdot 8} = 2^{24}$

f. $\left(\frac{21}{-4}\right)^5 =$
$\frac{21^5}{(-4)^5}$

4.1.2 REFRESHER: LAWS OF EXPONENTS

DIFFERENTIATE INSTRUCTION

Consider prompting students to highlight each exponent law and algebraic rule with a different color to help them distinguish between the laws.

ENRICHMENT

When a negative base is raised to an odd, positive exponent, what is the sign of the result? When a negative base is raised to an even, positive exponent, what is the sign of the result?

4.1.3 EXPLORATION: FINDING PATTERNS TO EVALUATE NUMERIC EXPRESSIONS WITH EXPONENTS

TEACHER MOVES

The 4.1.3 Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the 4.1.3 Exploration as they engage with each new reflection.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group during 4.1.4 Bring It Together. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.

4.1.3 Exploration: Finding Patterns to Evaluate Numeric Expressions with Exponents

Work with your partner to complete the following.

1. Complete the table.

Exponent Expression	Value
2^4	16
2^3	8
2^2	4
2^1	2
2^0	1

2. Describe any mathematical pattern(s) you notice regarding how the values of the expressions change as the exponent changes.
Answers will vary. As the exponent decreases, the value of the expression decreases. The value of the expression halves as the exponent decreases.

3. Complete the statement.
As the exponent (○ increases / ○ decreases) by 1, the value of the expression can be found by (○ multiplying / ○ dividing) the previous value by (○ 2. / ○ 4. / ○ 8.)

TEACHER MOVES

As you are monitoring student discussions on question 7 and responses to question 8, determine if any misconceptions need to be addressed or modifications need to be made during the 4.1.4 Bring It Together.

4. Continue the pattern to complete the table.

Exponent Expression	Value
2^1	2
2^0	1
2^{-1}	$\frac{1}{2^1} = \frac{1}{2}$
2^{-2}	$\frac{1}{2^2} = \frac{1}{4}$
2^{-3}	$\frac{1}{2^3} = \frac{1}{8}$

5. Compare the values of 2^3 and 2^{-3}. What do you notice?
$2^3 = 8$ $\quad$ $2^{-3} = \frac{1}{8}$
Answers will vary. The positive exponent has a whole number value, and the negative exponent is the inverse of the positive exponent's value.

6. Compare the values of 2^1 and 2^{-1}. What do you notice?

$2^1 = 2$	$2^{-1} = \frac{1}{2}$

Answers will vary. The positive exponent's value is a whole number and the negative exponent's value is a fraction.

7. With your partner, discuss how you can find the value of any number raised to a negative exponent. Summarize your discussion.
Answers will vary. When you are finding the value of any number raise to a negative exponent, take the inverse of the positive exponent value.

8. Based on the discussion with your partner, what is the value of 2^{-4}?
$2^{-4} = \frac{1}{2^4} = \frac{1}{16}$

4.1.4 Bring It Together: Negative Exponents

Expressions with **negative exponents** can be expressed with positive exponents: $x^{-a} = \frac{1}{x^a}$.

1. Complete the statement.
   A base with a negative exponent can be rewritten using the
   ○ reciprocal
   ○ opposite
   of the base with the
   ○ reciprocal
   ○ opposite
   exponent.

2. Complete the table.

Exponent Expression	Repeated Multiplication
a^3	$a \cdot a \cdot a$
a^2	$a \cdot a$
a^1	a
a^0	1
a^{-1}	$\frac{1}{a} = \frac{1}{a}$
a^{-2}	$\frac{1}{a} \cdot \frac{1}{a} = \frac{1}{a^2}$
a^{-3}	$\frac{1}{a} \cdot \frac{1}{a} \cdot \frac{1}{a} = \frac{1}{a^3}$

Negative Exponent Law

$a^{-1} = \frac{1}{a}$ and $\frac{1}{a} = a^{-1}$,

where a is any nonzero real number

4.1.4 BRING IT TOGETHER: NEGATIVE EXPONENTS

TEACHER MOVES

Students may confuse the concept of taking the reciprocal with finding the opposite of a number. Consider discussing the difference between these terms by showing students examples of each.

ENRICHMENT

Can we evaluate 0^{-2}? Why or why not?

4.1.5 GUIDED PRACTICE: REWRITING NEGATIVE EXPONENTS

DIFFERENTIATE INSTRUCTION

Use observational data from the 4.1.3 Exploration and 4.1.4 Bring It Together to determine how much guidance is needed to bring together key ideas in this 4.1.5 Guided Practice component. If necessary, consider modeling question 1 to address common misconceptions.

TEACHER MOVES

Students are not finding the value of each expression, only rewriting using positive exponents.

ENRICHMENT

Why is putting parentheses in the answer needed for question 3, but not for question 2?

TEACHER MOVES

Although it is not a specific vocabulary term, discussion of the term “reciprocal” may help students rewrite the expressions in questions 2, 4, and 6.

4.1.5 Guided Practice: Rewriting Negative Exponents

Rewrite each expression as an equivalent exponential expression using positive exponents.

1. $8^{-3} = \frac{1}{8^3}$
2. $\frac{1}{-4^{-5}} = \frac{-4^5}{1} = -4^5$
3. $(-2)^{-8} = \frac{1}{(-2)^8}$
4. $\frac{1}{10^{-2}} = \frac{10^2}{1} = 10^2$
5. $-5 \cdot 6^{-4} = -5 \cdot \frac{1}{6^4}$
6. $\frac{2^{-3}}{9^{-2}} = \frac{9^2}{2^3}$

4.1.6 Collaboration: Applying Other Exponent Laws with Negative Exponents

Work with your partner to complete the following.

1. Consider the exponential expressions in the table.

   a. For each expression, work with your partner to rewrite the exponential expression in expanded form and evaluate. The first row has been partially completed.

Exponential Expression	Apply Negative Exponent Law	Expanded Form	Value
$6^{-2} \cdot 6^5$	$\frac{1}{6^2} \cdot 6^5$	$\frac{1}{6 \cdot 6} \cdot 6 \cdot 6 \cdot 6 \cdot 6 \cdot 6 =$ $\frac{1}{6} \cdot \frac{1}{6} \cdot 6 \cdot 6 \cdot 6 \cdot 6 \cdot 6$	$\frac{1}{36} \cdot 7776$ $= 216$
$(-4)^3 \cdot (-4)^{-7}$	$-4^3 \cdot \frac{1}{-4^7}$	$-4 \cdot -4 \cdot -4 \cdot \frac{1}{-4 \cdot -4 \cdot -4 \cdot -4 \cdot -4 \cdot -4 \cdot -4}$	$\frac{-64}{-16384}$ $= \frac{1}{256}$
$3^{-4} \cdot 3^4$	$\frac{1}{3^4} \cdot 3^4$	$\frac{1}{3 \cdot 3 \cdot 3 \cdot 3} \cdot 3 \cdot 3 \cdot 3 \cdot 3$	$\frac{1}{81} \cdot 81$ $= 1$

4.1.6 COLLABORATION: APPLYING OTHER EXPONENT LAWS WITH NEGATIVE EXPONENTS

ENRICHMENT

How can you simplify the expanded form of the expressions? What is any number multiplied by its reciprocal equal to? How is multiplication by the reciprocal of a number related to dividing by the number?

4.1.6 COLLABORATION: APPLYING OTHER EXPONENT LAWS WITH NEGATIVE EXPONENTS (CONTINUED)

DIFFERENTIATE INSTRUCTION

As students progress through the lesson, consider prompting them to create a reference sheet or foldable with the laws of exponents (or if they have made one, add examples from the lesson as a reference). There are several laws reviewed, and students may benefit from seeing examples of each law on a reference sheet.

ENRICHMENT

How does the table in question 1b compare to the table in question 1a? Did any of the answers change? How were the processes different? Is there a method you prefer?

TEACHER MOVES

Consider allowing students to take turns sharing their thinking for each example. After each student shares, their partner can say if they agree or disagree and why.

TEACHER MOVES

This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.

Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.

b. Complete the table to evaluate each expression by applying the product of powers law.

Exponential Expression	Apply Product of Powers Exponent Law	Value
$6^{-2} \cdot 6^5$	$6^{-2+5} = 6^{\boxed{3}}$	216
$(-4)^3 \cdot (-4)^{-7}$	$(-4)^{3-7} = (-4)^{-4}$	$\frac{1}{256}$
$3^{-4} \cdot 3^4$	$3^{-4+4} = 3^0$	1

c. With your partner, discuss your observations of the tables from Parts A and B.
Answers will vary. The negative exponent law ensures that all exponents are positive. When finding the values using this law, we multiply exponents. When finding values using the product of powers law, we add the exponents.

2. Consider the exponential expressions in the table.

a. For each expression, work with your partner to rewrite the exponential expression in expanded form and evaluate. The first row has been partially completed.

Exponential Expression	Apply Negative Exponent Law	Expanded Form	Value
$(-8)^{-4} \div (-8)^{-6}$	$\frac{1}{(-8)^4} \cdot (-8)^6$	$\frac{(-8)(-8)(-8)(-8)(-8)(-8)}{(-8)(-8)(-8)(-8)}$	64
$7^0 \div 7^{-4}$	$\frac{1}{7^0} \cdot \frac{7^4}{1}$	$\frac{2401}{1}$	2401
$9^{-5} \div 9^3$	$\frac{1}{9^5} \cdot \frac{1}{9^3}$	$\frac{1}{9 \cdot 9 \cdot 9 \cdot 9 \cdot 9 \cdot 9 \cdot 9 \cdot 9}$	$\frac{1}{43046721}$

b. Complete the table to evaluate each expression by applying quotient of powers law.

Exponential Expression	Apply Quotient of Powers Exponent Law	Value
$(-8)^{-4} \div (-8)^{-6}$	$(-8)^{-4-(-6)} = (-8)^{\boxed{2}}$	64
$7^0 \div 7^{-4}$	$7^{0-(-4)} = 7^4$	2401
$9^{-5} \div 9^3$	$9^{-5-3} = 9^{-8}$	$\frac{1}{43046721}$

c. With your partner, discuss your observations of the tables after completing Parts A and B.
Answers will vary. Negative exponents represent reciprocals. When applying the quotient of powers exponent law, the exponents are subtracted; use parentheses when the second exponent is negative.

3. Several equivalent expressions are shown using the negative exponent law.

$$(5^{-3})^2 = \left(\frac{1}{5^3}\right)^2 = \left(\frac{1}{5 \cdot 5 \cdot 5} \cdot \frac{1}{5 \cdot 5 \cdot 5}\right) = \frac{1}{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5} = 5^{-6}$$

Which exponent law(s) can also be used to show $(5^{-3})^2 = 5^{-6}$?
Power of a Power Rule

4. Complete the statement.

The laws of exponents ○ do / ○ do not hold true for numeric expressions with rational bases and negative exponents.

4.1.7 Wrap-Up: Growth Mindset

Complete each statement based on today's lesson.

1. Today I am still unsure about …
   Answers will vary.

2. I will try to build my confidence by …
   Answers will vary.

4.1.7 WRAP-UP: GROWTH MINDSET

DIFFERENTIATE INSTRUCTION

The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

